

Minutes of The Third Meeting

OS1 group B

20th Mar 2025

Chair Orvila Sarker

Secretary Sheng Wang

Members Cong Deng

Tianhua Zhang

Xin Wei

Yifan Gu

Apologies None.

1. Time and Place

The third team meeting for the MCI OS1 project was held at **Inggarni Wardli 5.57** on **20th Mar** at **3:30 PM**.

2. Quorum Announcement

The Chairman announced that a quorum of the group was present, and that the meeting, having been duly convened, was ready to proceed with its business.

3. Summary of Previous Meeting

The team has defined the project's overarching objective, technical research direction, and specific tasks, along with their respective work deadlines, and established an overview of the project plan.

4. Group Milestone

4.1. Overview

The team provided the supervisor with an update on the progress, challenges, and implementation details of Task 2. The discussion covered key questions, proposed suggestions, and outlined the next steps for the task. Additionally, the supervisor provided guidance on the scope and key considerations for the Week 4 pitch presentation.

4.2. Task Division and Completion

The team has divided Task 2 into the following key tasks:

4.2.1. Manual Annotation of Security-Related Questions

- a. A total of 386 security-related questions were manually labeled and categorized into different types(debugging, how-to and conceptual).
- b. To ensure labeling accuracy and reduce bias, two members independently labeled questions 0-194(Xin Wei, Sheng Wang), while another two members independently labeled questions 194-386(Cong Deng, Yifan Gu).
- c. One team member manually retrieved the view count for each post to ensure data accuracy(Tianhua Zhang).

4.2.2. Model Selection and Implementation

- a. The entire team participated in discussions regarding model selection and the initial implementation(All Team).
- b. Based on the discussion outcomes:
 - Two team members(Tianhua Zhang, Cong Deng)were responsible for the main coding tasks, focusing on SVM and Random Forest implementations.
 - Three team members (Yifan Gu, Sheng Wang, and Xin Wei) were responsible for the core coding tasks, with a focus on implementing LDA (Latent Dirichlet Allocation) and BERT (Bidirectional Encoder Representations from Transformers) models.

All Team members conducted code reviews, model training, analysis of training

results, and validation of model accuracy using publicly available datasets.

4.3. Challenges Faced

4.3.1. Data Cleaning

Model accuracy depends on data cleaning and labeling, but security data's complexity (diverse terms, languages) challenges effective preprocessing.

4.3.2. Model Selection

Choosing the appropriate models (SVM, LDA, BERT) and training approaches (supervised, unsupervised, semi-supervised) and determining the optimal combination for this project requires extensive data validation and team discussions.

5. Submission

5.1. Data Labeling

Categorical labeling of 386 security-related issues into predefined categories.

5.2. Model Development and Analysis

Implementation of SVM, LDA and BERTopic model, along with aggregation, comparison, and review of preliminary data results.

6. Next Planing

6.1. mode

Supervised Learning: Continuously optimize the classification accuracy of the SVM model.

- Unsupervised Learning: Compile a comparative report on different unsupervised learning models (LDA, Bert-Topic) and their performance.
- Team Discussion: Conduct a group discussion to review and analyze the model results.

6.2. Pitch Presentation Planning

The team's Pitch Presentation is scheduled in the fourth week on **27th Mar**. The presentation will focus on the project's key selling points and features, as well as the team's current progress on Task 1 and Task 2, and the next steps for the project.

7. Adjournment

The next team meeting is scheduled for **27th Mar** at **3:30 PM**.